

Processando config 29, seed 4, teste: Rabbit
 xx
 Starting Rabbit: nb = 16777216
 Version: TestU01 1.2.3
 xx

 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

smultin_MultinomialBitsOver test:

 N = 2, n = 8388608, r = 0, s = 32, L = 38,
 Sparse = TRUE

Number of bits = n = 8388608
 Number of cells = 2^L = 274877906944
 Expected number per cell = $1 / 32768$
 $E_{\text{Coll}} = n^2 / (2k) = 128$
 Hashing = TRUE

Collision test
 CollisionOver: density = $n / k = 1 / 32768.00$
 Expected number of collisions = $\mu = 128.00$

 Results of CollisionOver test:

POISSON approximation :
 Expected number of collisions = $N \cdot \mu$: 256.00
 Observed number of collisions : 224
 p-value of test : 0.98

 Total number of cells containing j balls

j = 0	:	549739036896
j = 1	:	16776768
j = 2	:	224
j = 3	:	0
j = 4	:	0
j = 5	:	0

 CPU time used : 00:00:04.50

Generator state:
 16777216 bits have been read.

 HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.04

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.08

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -0.65
p-value of test : 0.74

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 6.42
p-value of test : 0.49

Normal statistic for number of jumps : -51.39
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.05

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 1.27
p-value of test : 0.10

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -1.45
p-value of test : 0.93

CPU time used : 00:00:00.20

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.019
p-value of test : 0.69

Kolmogorov-Smirnov- statistic = D- : 0.018
p-value of test : 0.70

Anderson-Darling statistic = A2 : 0.32
p-value of test : 0.93

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 3.51
p-value of test : 0.32

Global longest run of 1 : 23.00
p-value of test : 0.50

CPU time used : 00:00:00.07

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 83.66
p-value of test : 0.77

CPU time used : 00:00:00.01

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 16.23
p-value of test : 0.80

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.012
p-value of test : 0.50

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.75
p-value of test : 0.23

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.92
p-value of test : 0.18

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 159.91
p-value of test : 0.82

CPU time used : 00:00:00.01

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 317.11
p-value of test : 0.30

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 537.43
p-value of test : 0.06

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1


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-----
Normal statistic          :    1.01
p-value of test          :    0.16
-----
```

```
CPU time used            : 00:00:00.01
```

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Generator state:
269499904 bits have been read.
```

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*****
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin
```

```
sstring_AutoCor test:
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```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :   -0.75
p-value of test          :    0.77
-----
```

```
CPU time used            : 00:00:00.00
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```
Generator state:
286277088 bits have been read.
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*****
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin
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sstring_Run test:
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-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355449
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 31.66
p-value of test : 0.67
-----
```

```
-----
Total number of bits: 13420416
```

```
Normal statistic for number of bits : -0.26
p-value of test : 0.60
```

CPU time used : 00:00:00.04

Generator state:
299697504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 0.37
p-value of test : 0.95

CPU time used : 00:00:00.07

Generator state:
316474720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333165920 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	34.77
p-value of test	:	0.78

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	55.47
p-value of test	:	0.08

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	72.38
p-value of test	:	0.22

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	23.86
p-value of test	:	0.94

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	10.36
p-value of test	:	0.98

CPU time used : 00:00:00.04

Generator state:
349943136 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	96.05
p-value of test	:	0.42

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	93.57
p-value of test	:	0.49

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	510.25
p-value of test	:	0.51

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	101.47
p-value of test	:	0.19

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	65.28
p-value of test	:	0.07

CPU time used : 00:00:00.04

Generator state:

366720352 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	114.26
p-value of test	:	0.84

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	135.76
p-value of test	:	0.37

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	159.75
p-value of test	:	0.51

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	108.89
p-value of test	:	0.91

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	85.38
p-value of test	:	0.59

CPU time used : 00:00:00.03

Generator state:

383497152 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:05.98
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
5 LinearComp	1 - eps1
22 MatrixRank, 320 x 320	eps
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All other tests were passed